

Pentose Phosphate Pathway

HMP Shunt

Hexose Mono Phosphate Shunt = Pentose Phosphate Pathway = or 6-phosphogluconate pathway

- It consists of two, irreversible oxidative reactions, followed by a series of reversible sugar-phosphate interconversions.
- No ATP is directly consumed or produced in the cycle.

Function : Production of

- For NADPH
- Ribose 5P

Site :

- In the cytoplasm of all cells except muscle, and nonlactating mammary gland (low activity)

Why is the pentose phosphate pathway necessary?

- ATP is the “energy currency” of cells, but cells also need **reducing power**.
- Endergonic reactions require reducing power and ATP
 - Fatty acids, cholesterol, photosynthesis
- **NADPH and NADH are not interchangeable!**
 - Differ only by a phosphate group at the 2'OH.

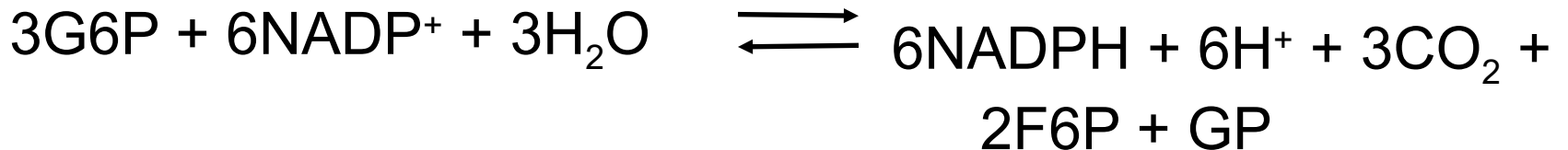
Pentose phosphate pathway

- **NADPH and NADH are not interchangeable!**
 - Differ only by a phosphate group at the 2'OH.
- **NADH** participates in utilizing the free energy of metabolite oxidation to synthesize ATP
- **NADPH** utilizes the free energy of metabolite oxidation for biosynthesis
- Difference is possible because the dehydrogenase enzymes involved in oxidative and reductive metabolism exhibit a high degree of specificity toward their respective coenzymes.
- **Ratios different:**
 - **[NAD⁺]/[NADH] is near 1000** which favors metabolite oxidation.
 - **[NADP⁺]/[NADPH] is near 0.1** which favors metabolite reduction.

Why is the pentose phosphate pathway necessary?

- **NADPH** is generated by **oxidation of G6P** via the **pentose phosphate pathway**
 - hexose monophosphate (HMP) pathway, phosphogluconate pathway.
- Alternate to glycolysis.
- Produces **ribose-5-phosphate** (essential for nucleotide biosynthesis).

Overall reaction



Can be considered in 3 stages

- Generation of NADPH
 - mainly used for reductive syntheses of fatty acids, steroids, amino acids via glutamate dehydrogenase; and production of reduced glutathione in erythrocytes and other cells.
 - active in liver, adipose tissue, adrenal cortex, thyroid, erythrocytes, testes, and lactating mammary gland
 - not active in non-lactating mammary gland and has low activity in skeletal muscle.
- **Production of ribose** residues for nucleotide and nucleic acid synthesis.

Characteristics: Tissue Distribution

- Demand for NADPH
 - Biosynthetic pathways
 - FA synthesis (liver, adipose, mammary)
 - Cholesterol synthesis (liver)
 - Steroid hormone synthesis (adrenal, ovaries, testes)
 - Detoxification (Cytochrome P-450 System) – liver
 - Reduced glutathione as an antioxidant (RBC)
 - Generation of superoxide (neutrophils)

NADPH for H₂O₂ elimination

- In the Erythrocytes, Pulmonary Cells, and Liver Cells :



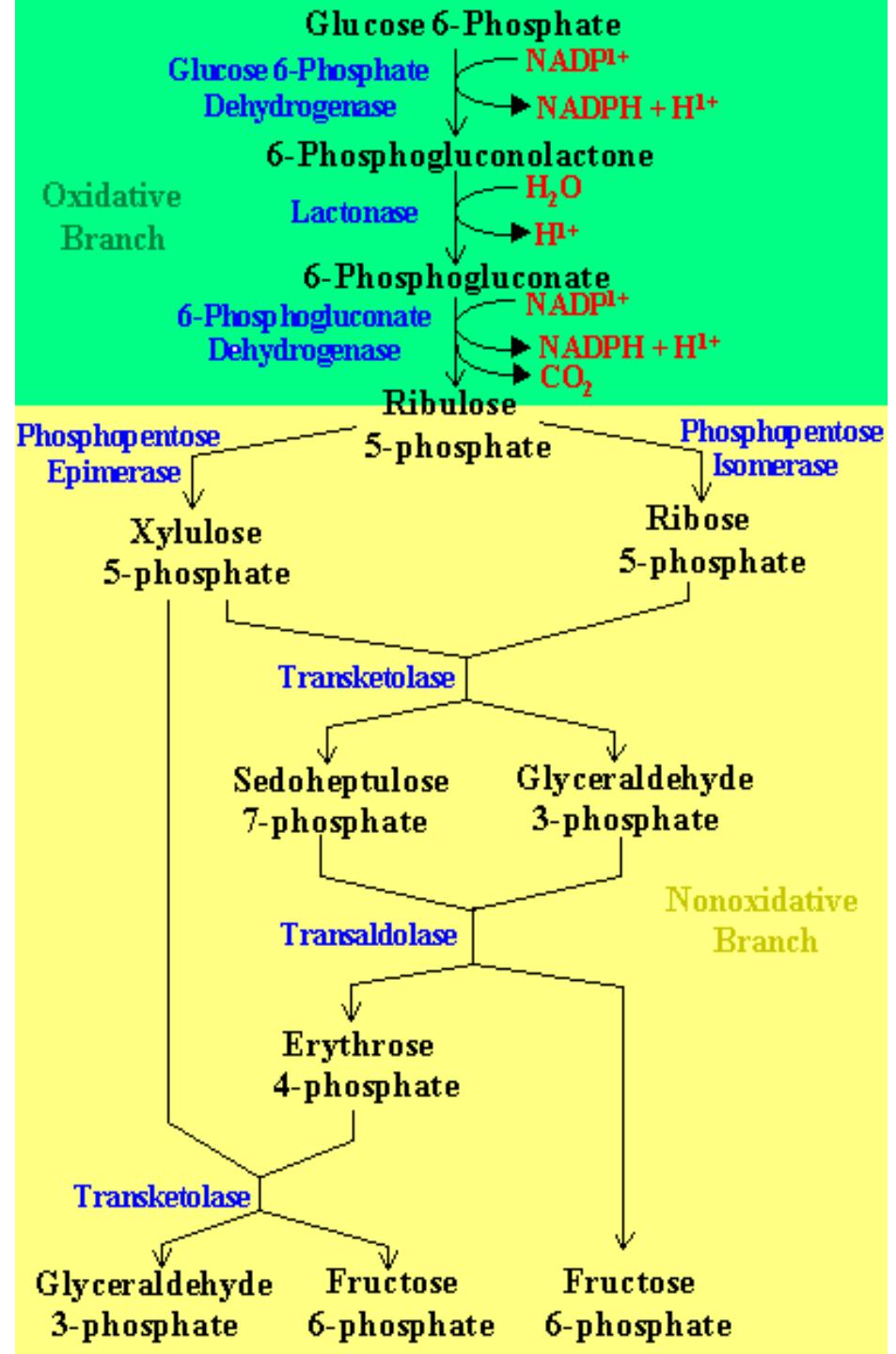
Enzyme 1. Glutathione peroxidase

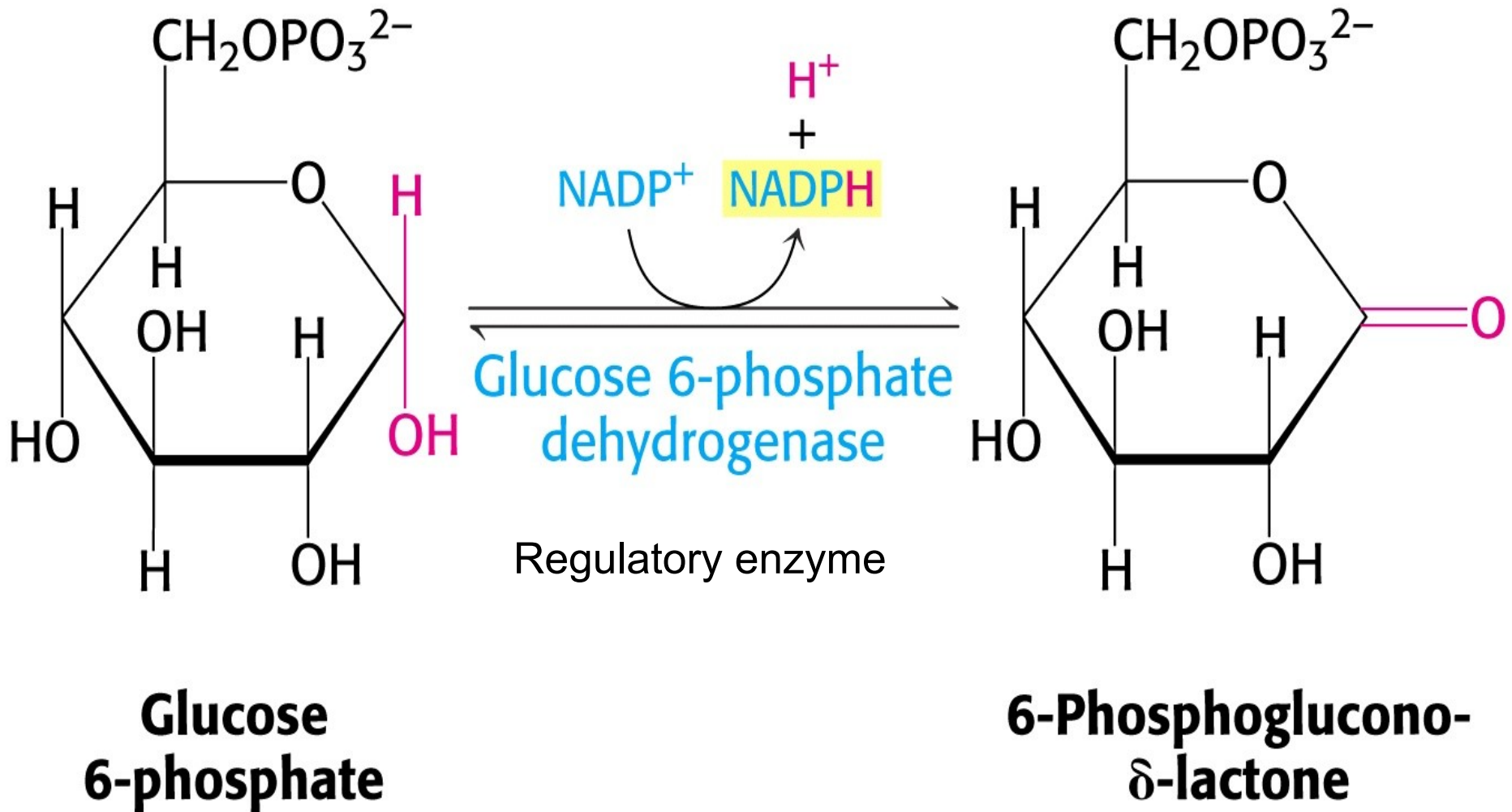
Enzyme 2. Glutathione reductase

NADPH + H⁺ is formed from two separate reactions.

The glucose 6-phosphate DH (G6PD) reaction is the rate limiting step and is essentially irreversible.

Cells have a greater need for NADPH than ribose 5-phosphate.

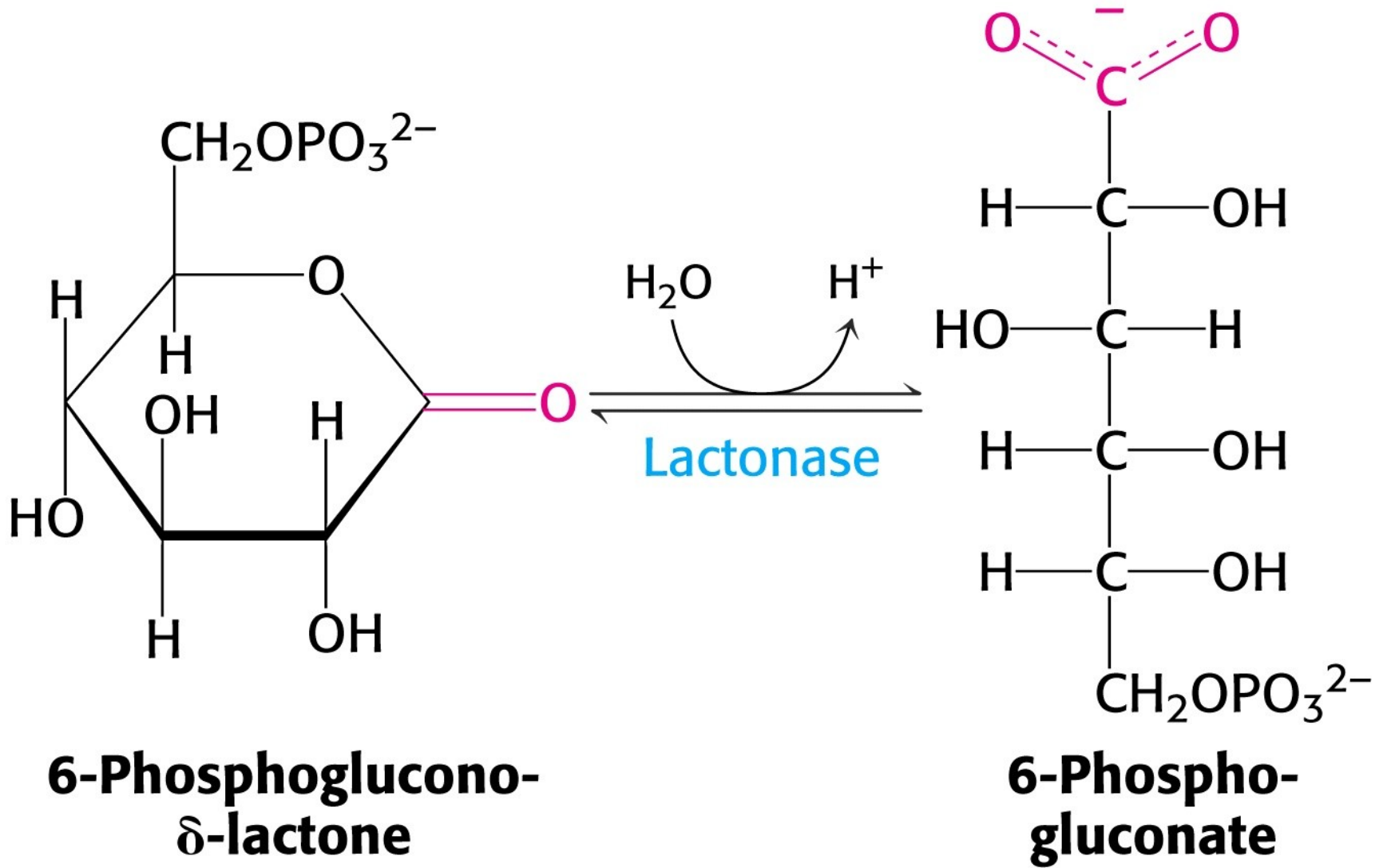


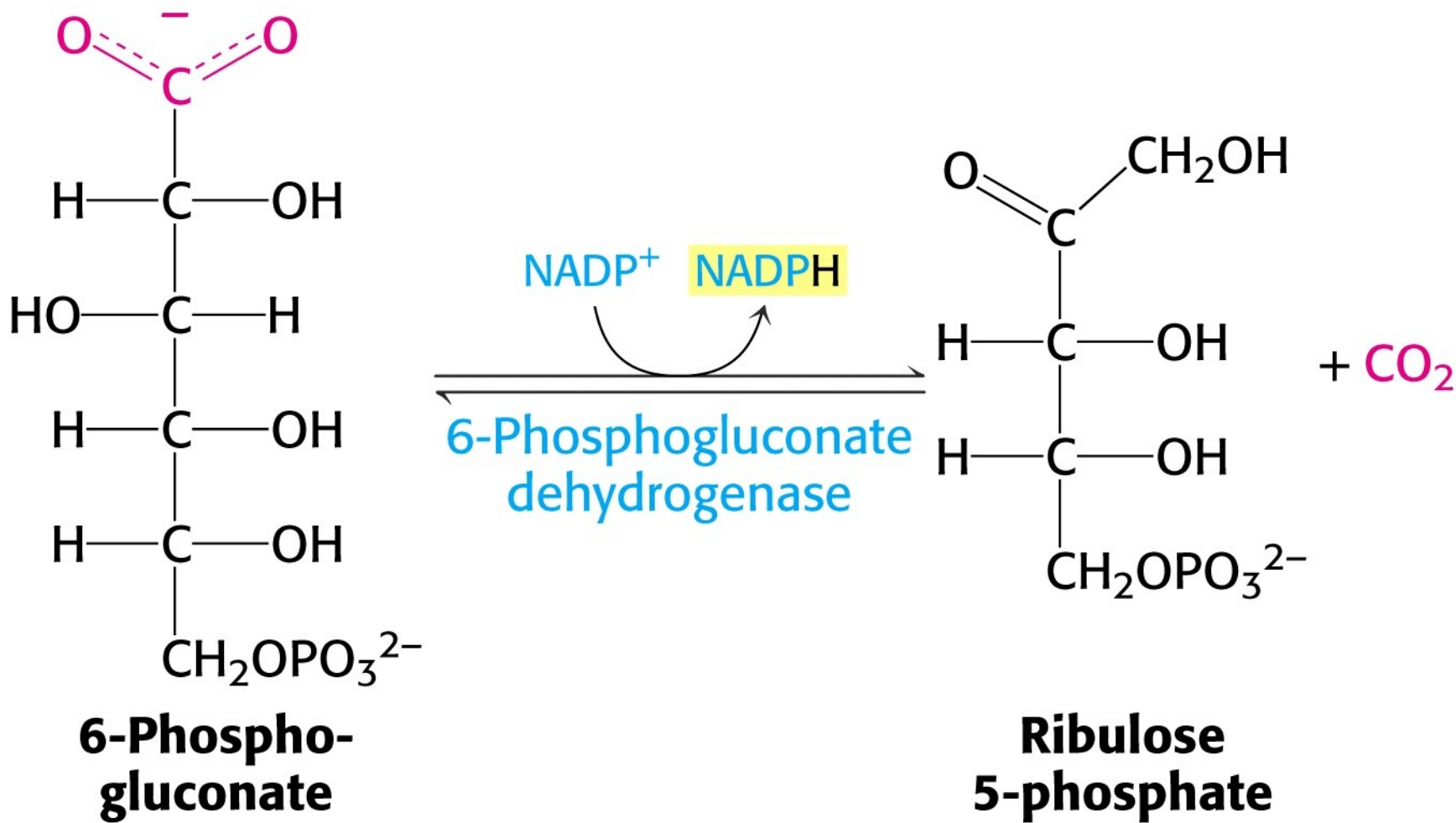


The enzyme is highly specific for NADP^+ ; the K_m for NAD^+ is 1000 greater than for NADP^+ .

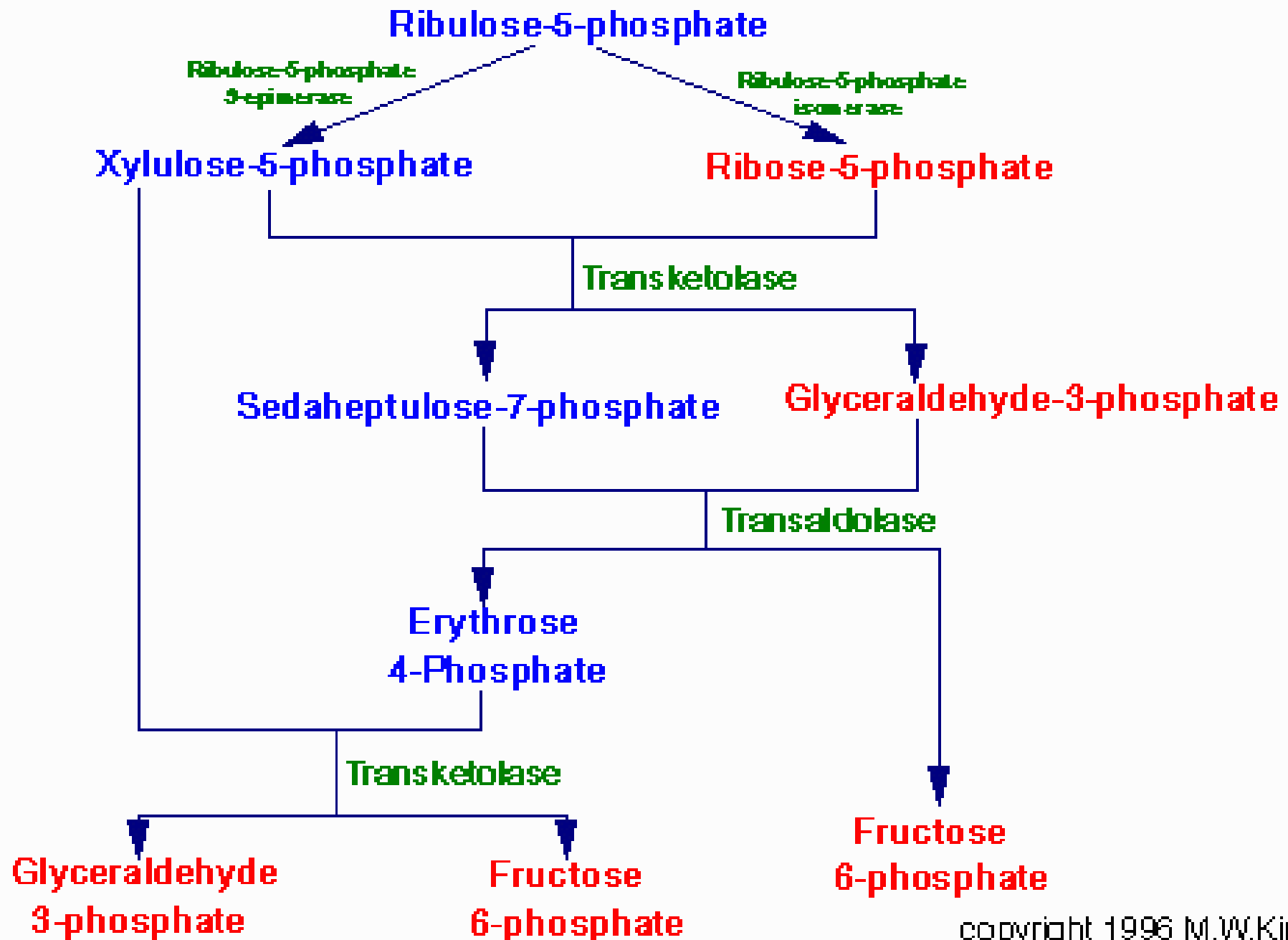
Regulation

- Glucose-6-P dehydrogenase
 - First step
 - Rate limiting
- Allosteric Regulation
 - Feedback inhibited by NADPH
- Inducible enzyme
 - Induced by insulin



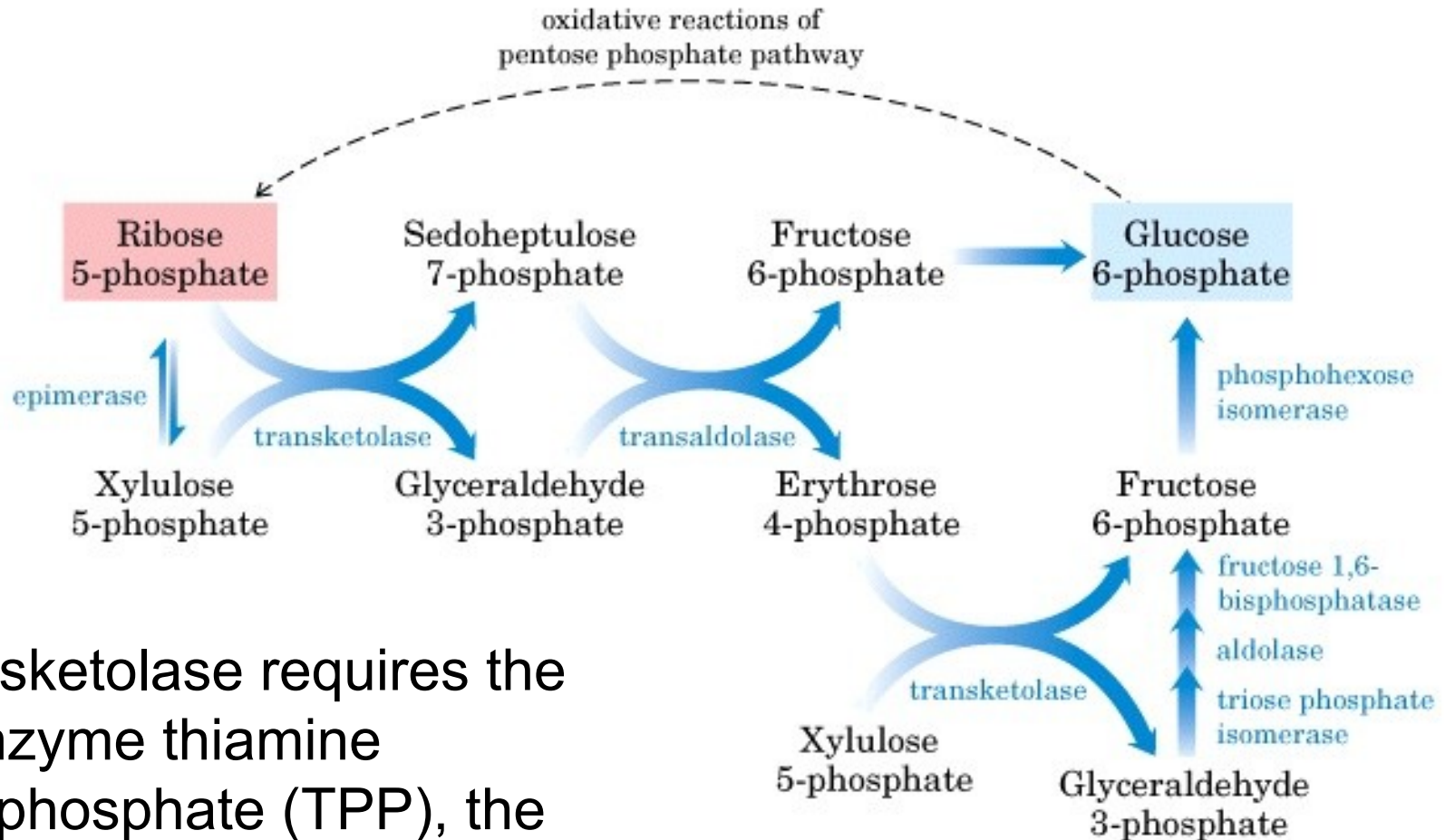


Non-Oxidative Stage of Pentose Phosphate Pathway



The nonoxidative phase of the pentose phosphate pathway

This entails extensive carbon atom rearrangement.



Transketolase requires the coenzyme thiamine pyrophosphate (TPP), the transaldolase does not.

(a)

- Transketolase (TPP) and transaldolase are the link back to glycolysis.
- Glyceraldehyde 3-phosphate
- Fructose 6-phosphate
- Net result:



Control of Pentose Phosphate Pathway

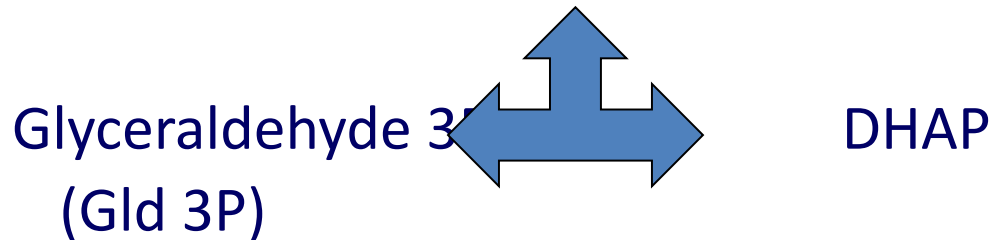
1. Principle products are R5P and NADPH.
2. Transaldolase and transketolase convert excess R5P into glycolytic intermediates when NADPH needs are higher than the need for nucleotide biosynthesis.
3. GAP and F6P can be consumed through glycolysis and oxidative phosphorylation.
4. Can also be used for gluconeogenesis to form G6P
5. 1 molecule of G6P can be converted via 6 cycles of PPP and gluconeogenesis to 6 CO₂ molecules and generate 12 NADPH molecules.
6. Flux through PPP (rate of NADPH production) is controlled by the glucose-6-phosphate dehydrogenase reaction.
7. G6PDH catalyzes the first committed step of the PPP.

In the muscle

- HMP Shunt inactive because G 6P Dehydrogenase and 6 P Gluconate Dehydrogenase deficient
- Ribose 5 P synthesized in the way of reverse HMP Shunt or through Transketolase path.

Synthesis Ribose 5P in the muscle

$G \rightarrow G\ 6P \rightarrow F\ 6P \rightarrow F\ 1,6\ BP$



$Gld\ 3P + F\ 6P \rightarrow Xylulose\ 5P + Erythrose\ 4P$

$Xylulose\ 5P \rightarrow Riboluse5\ P \rightarrow Ribose\ 5P$

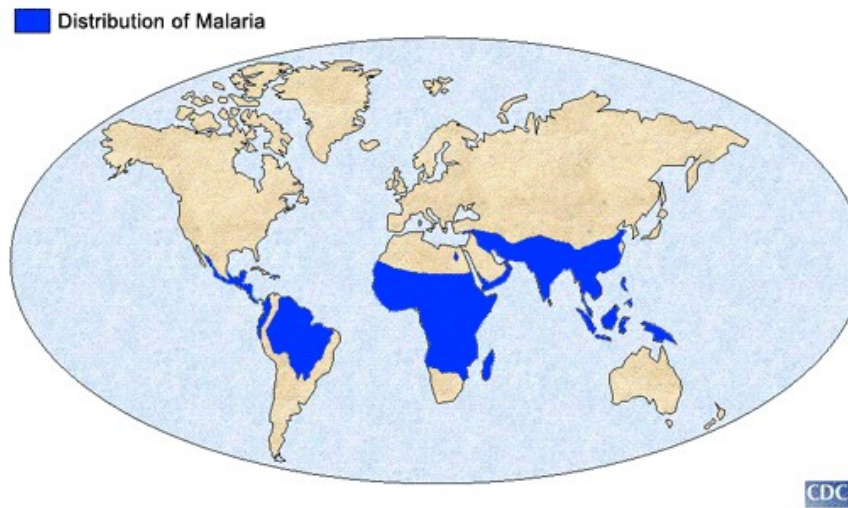
Glucose-6-phosphate dehydrogenase (G6PD) deficiency causes hemolytic anemia

- Mutations present in some populations causes a deficiency in glucose 6-phosphate dehydrogenase, with consequent impairment of NADPH production.
- Detoxification of H_2O_2 is inhibited, and cellular damage results - lipid peroxidation leads to erythrocyte membrane breakdown and hemolytic anemia.
- Most G6PD-deficient individuals are asymptomatic - only in combination with certain environmental factors (sulfa antibiotics, herbicides, antimalarials, *divicine) do clinical manifestations occur.

*toxic ingredient of fava beans

G6PD Deficiency

- Distribution of G6PD deficiency coincides prevalence of malaria



- G6PD deficiency may impart some degree of malaria resistance
 - Also sickle cell anemia

Genetics

- Recessive sex-linked mutation
 - X-chromosome
 - Rare in females (two X-chromosomes)
- Homozygous mutation:
 - high hemolysis and anemia
- Heterozygous mutation:
 - Normally asymptomatic
 - unless exposed to drugs (primaquine, anti-malarial drug) or compounds (fava bean) that produce superoxide or hydrogen peroxide

G6PD Deficiency

- Exposure to anti-malarial drugs (Primaquine) results in increased cellular production of superoxide and hydrogen peroxide (Primaquine sensitivity)
- Other chemicals known to increase oxidant stress
 - Sulfonamides (antibiotic)
 - Asprin and NSAIDs
 - Quinadine and quinine
 - Naphthlane (mothballs)
 - Fava beans (vicine & isouramil)

Fava Beans

- Grown worldwide
 - Important in Middle East
 - High in protein
 - Frost resistant perennial
- Genetically modified fava bean being developed
 - Low in vicine and isouramil
- Favism



Glucuronic acid:

- Glucuronic acid can be obtained in small amounts from diet. It can also be obtained from the intracellular lysosomal degradation of GAGs, or via the uronic acid pathway.
- The end-product of glucuronic acid metabolism in humans is D-xylulose 5-P, which can enter the hexose monophosphate pathway & produce the glycolytic intermediates GA-3P & F-6-P
- The active form of glucuronic acid that donates the sugar in GAG synthesis & other glucuronylating reactions is UDP-glucuronic acid, which is produced by oxidation of UDP-glucose

URONIC ACID PATHWAY